I/O Devices

Snake n' Shake — Hardware Interface Overview

Camera	VGA Monitor	7-Seg Display	Push Buttons
Player Input	Game Output	Score Output	Controls / Debug

Input & Output Devices Flowchart

Camera

FPGA

VGA Monitor

Input & Output Devices

Camera

Device	D5M Camera (CMOS Sensor)
Interface	Parallel + SCCB (I2C)
Purpose	Player input via color detection
Protocol	I2C Master RTL for config
Output	8-bit Pixel Data (PCLK)
Role	Feeds SDRAM frame buffer

VGA Monitor

Device	VGA Monitor / Display
Interface	VGA (analog signal)
Purpose	Game output to screen
Resolution	640×480 @ 60 Hz
Controller	VGA Controller RTL
Role	Renders game frame data

Control & Feedback Devices

7-Segment Display

Device	DE1-SoC 7-Segment Display
Interface	GPIO (on-board)
Purpose	Score output during gameplay
Displays	Current score & high score
Type	Hardware-driven, no CPU
Role	Live in-game feedback

Push Buttons & Switches

Device	DE1-SoC Push Buttons
Interface	GPIO (on-board)
Purpose	Reset, pause, movement
Fallback	Backup if camera fails
Switches	Sensitivity threshold tuning
Role	Debug & fallback controls